MOBILE COMPUTING LABORATORY

EXPERIMENT 04

Key Elements of Mobile Network Protocol Stacks

A layered view of how mobile network communication moves from user-facing services to radio transmission.

LAYERED ARCHITECTURE

Mobile Protocol Stack

APPLICATION ↓ MEDIUM

05

LAYER 05

Application Layer

Provides network services directly to user applications.

HTTP FTP DNS

04

LAYER 04

Transport Layer

Manages end-to-end delivery, reliability, and data flow.

TCP UDP

03

LAYER 03

Network Layer

Handles logical addressing and routes packets across networks.

IP

02

LAYER 02

Data Link Layer

Organizes frames and controls access to the wireless medium.

MAC LLC

01

LAYER 01

Physical Layer

Transmits raw data as radio waves through the physical medium.

WIRELESS SIGNALS

RADIO TRANSMISSION

Each layer serves the layer above it and communicates with its corresponding layer on the receiving device.

LAB RECORD • LAYERED DIAGRAM

MOBILE NETWORK PROTOCOLS